

Module 3: MAC Sublayer & LANs

Pure & Slotted ALOHA Proofs, CSMA/CD Minimum Frame Derivation, Exponential Backoff, Ethernet & Wi-Fi

☰ Module III: Error Handling, Data Link Control & Multiplexing — Complete Tracker

- Topic 20:** Types of Errors (Single-Bit Errors vs. Burst Errors)
- Topic 21:** Error Detection Techniques (Simple Parity & Limitations)
- Topic 22:** Checksum (1's Complement Summation & Verification)
- Topic 23:** Cyclic Redundancy Check (CRC-32 Modulo-2 Division)
- Topic 24:** Error Correction (Hamming Code & Syndrome Decoding)
- Topic 25 & 26:** Flow & Error Control ARQ (Stop-and-Wait, GBN, Selective Repeat)
- Topic 27:** High-Level Data Link Control (HDLC Frame: I, S, U-Frames)
- Topic 28 & 29:** Multiplexing (FDM, Synchronous TDM vs. Statistical TDM)

Topic 20 – 22: Error Types & Basic Detection Techniques

Error Class	Mathematical Definition	Detection Feasibility
1. Single-Bit Error	Only 1 bit in the data block is inverted from 0 → 1 or 1 → 0.	Easily detected by a single Parity Bit.
2. Burst Error	Two or more bits in a sequence are corrupted. Length of burst is measured from first corrupted bit to last corrupted bit.	Requires Cyclic Redundancy Check (CRC) or 2D Parity.

Topic 23: Cyclic Redundancy Check (CRC) Polynomial Division [UPLOADED PYQ]

 [UPLOADED PYQ] Solved Problem: CRC for Data `1010011110` with Generator $G = 1011$

Step 1: Determine Degree of Generator: $G = 1011$ has degree $r = 3$.

Step 2: Append $r = 3$ zeros to Data: Augmented Dividend = `1010011110000`.

Step 3: Modulo-2 XOR Division:

```
      1001001011 (Quotient)
1011 ) 1010011110000
      1011
      ---
      001011
       1011
       ---
       00001110
        1011
        ---
        1010
        1011
        ---
        001000
         1011
         ---
         0110 ← Remainder CRC = 011
```

Step 4: Transmitted Frame: Data + CRC = 1010011110011.

Topic 24: Error Correction (Hamming Code (n, k) Derivation)

Hamming Parity Bit Invariants (Hamming, 1950): For m data bits and r parity check bits:

$$2^r \geq m + r + 1$$

- Parity bits are placed strictly at bit positions that are powers of 2 (1, 2, 4, 8, 16, ...). - Data bits occupy all remaining positions (3, 5, 6, 7, 9, 10, ...). - Parity bit P_k checks all bit positions whose binary representation has a 1 in the k -th position. - **Syndrome:** Evaluation of parity checks at the receiver forms a binary integer: 000 $\implies$ No error; non-zero value indicates the exact corrupted bit position!

Topic 25 & 26: Sliding Window Flow & Error Control ARQ Protocols [UPLOADED PYQ]

ARQ Protocol	Sender Window (W_s)	Receiver Window (W_r)	Key Retransmission Behavior & Efficiency
1. Stop-and-Wait ARQ	$W_s = 1$	$W_r = 1$	Sender waits for ACK before sending next frame. Low efficiency on long propagation delays ($\eta = \frac{1}{1+2a}$).
2. Go-Back-N (GBN) [UPLOADED PYQ]	$W_s \leq 2^m - 1$	$W_r = 1$ (Cumulative ACK)	When frame k is lost or damaged, receiver discards all subsequent out-of-order frames. Sender retransmits all frames from k onwards.
3. Selective Repeat (SR)	$W_s \leq 2^{m-1}$	$W_r \leq 2^{m-1}$ (Independent ACK)	Receiver buffers out-of-order frames. Sender retransmits <i>only the specific missing frame k</i> , saving bandwidth.

Topic 27: High-Level Data Link Control (HDLC Frame Formats)

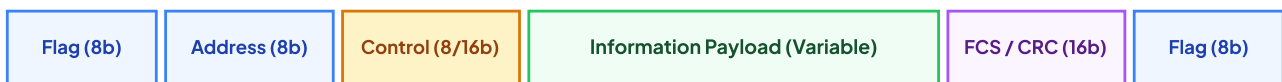


Figure 3.1: Universal HDLC Frame Architecture with Flag `01111110` Delimiters

The 3 HDLC Frame Formats

- 1. **Information Frames (I-Frames)**: Carries user data payload and piggybacked ACKs ($N(S)$ send sequence and $N(R)$ receive sequence).
- 2. **Supervisory Frames (S-Frames)**: Carries flow/error control without payload (RR - Receive Ready, RNR - Receive Not Ready, REJ - Reject / NAK).
- 3. **Unnumbered Frames (U-Frames)**: Used for link initialization, mode setting (SABM - Set Asynchronous Balanced Mode), and link disconnection (DISC).

Topic 28 & 29: Multiplexing (FDM, Synchronous vs. Statistical TDM) [UPLOADED PYQ]

Feature	Synchronous TDM	Statistical / Asynchronous TDM [UPLOADED PYQ]
Time Slot Allocation	Pre-assigned, fixed, dedicated time slots to each input line in a round-robin cycle.	Time slots allocated dynamically on-demand only to active input lines with data to send.
Bandwidth Efficiency	Low: if an input line has no data to transmit, its allocated time slot goes completely empty/wasted.	High: zero wasted empty slots. Total input data rate can exceed the aggregate multiplexed link capacity.
Addressing Overhead	No addressing needed in slot (receiver knows source by slot position).	Each slot requires header addressing metadata to identify source/destination.



M3 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Differentiate between Stop-and-Wait ARQ and Go-Back-N ARQ. (8 Marks)

1. **Sender Window Size:** Stop-and-Wait has $W_s = 1$; Go-Back-N has $W_s = 2^m - 1 > 1$, enabling continuous transmission pipelining.
2. **Channel Utilization:** Stop-and-Wait has very poor link utilization ($\eta = \frac{1}{1+2a}$); GBN achieves near 100% utilization when $W_s \geq 1 + 2a$.
3. **Retransmission Scope:** In Stop-and-Wait, only 1 frame is retransmitted on timeout; in GBN, the entire window of N unacknowledged frames is retransmitted upon a single lost frame.